

Hospital Management System

System Overview, Technology Stack, and User Guide

System Overview

This Hospital Management System (HMS) is a web-based application for managing patient flow, clinical work, pharmacy operations, and administrative oversight. It supports multiple roles (admin, receptionist, nurse, doctor, pharmacist) and integrates realtime updates for queue and operational events.

High-Level Architecture

The system is split into three main layers:

- Frontend: Static HTML/JS pages under the frontend/ directory. These pages call backend APIs and render role-specific dashboards.
- Backend API: PHP endpoints routed via backend/index.php. Each module (admin, reception, doctor, nurse, pharmacy, etc.) has a dedicated route file.
- Realtime Server: A Node.js WebSocket service in realtime/server.js used to broadcast events to connected clients.

Technology Stack

Core technologies used in this project:

- Frontend: HTML, CSS, JavaScript with Bootstrap 5.3.3 and Bootstrap Icons (CDN references).
- Backend: PHP (API routing and controllers).
- Authentication: JWT tokens via firebase/php-jwt (Composer dependency).
- Database: MySQL (schema in backend/config/database/hospital_db.sql).
- Realtime: Node.js with ws WebSocket library (realtime/package.json, realtime/server.js).
- Hosting/Local Stack: Works in XAMPP or similar Apache + MySQL setup.

Security and Sessions

The system uses JWT access tokens for API calls. Tokens include a session id and expiration. The backend enforces role-based access via middleware and limits each account to a maximum of three active sessions.

Key Data Domains

Primary database areas (not exhaustive):

- Users and roles (users), including session tracking.
- Patients, queue, and vitals (patients, patient_queue, patient_vitals).
- Appointments and visit tracking (appointments, patient_queue).
- Prescriptions and dispensing (prescriptions, medicines, controlled_substances).
- Pharmacy operations (refill_requests, suppliers, purchase_orders, goods_receipts, insurance_claims).
- Operational logs (activity_logs and audit_logs).
- Staff shifts and handovers (staff_shifts, doctor_handover, nurse_handover, reception_handover).
- Documents and reports (medical_reports, uploads).

User Guide

1. Logging In

- Open the login page and enter your username and password.
- If you already have three active sessions, the system will block the new login until you log out of another session.

2. Navigation

- After login, you are routed to the dashboard for your role (Admin, Reception, Nurse, Doctor, Pharmacy).
- Use the sidebar or tabs to access module pages such as patients, queue, reports, and settings.

3. Common Tasks

- Search for patients by name, ID, or phone from Reception or clinical pages.
- Use queue tools to admit patients, move them through triage, and complete visits.
- Attach or upload documents to a patient record when required.

Role-Specific Workflows

Admin

- View system stats and operational metrics on the admin dashboard.
- Create and manage staff accounts (users).
- Review system logs and audit activity.
- Configure settings such as maintenance mode.
- Manage staff shifts and roster data.

Reception

- Register new patients and update demographic details.
- Admit patients into the queue and assign a doctor when appropriate.
- Create and manage appointments, and check in scheduled patients.
- Manage queue status (cancel, reassign, move to top).
- Upload patient documents and create referral or consent records.
- Record shift handover notes.

Nurse

- View triage queue and start triage for waiting patients.
- Record vitals and route patients to the doctor queue.
- Manage beds, assign beds, and discharge patients.
- Create nurse tasks, escalations, and discharge summaries.
- Review patient history and trends.

Doctor

- View the doctor waiting list after triage completion.
- Document consultation notes and complete visits.
- Create prescriptions (including external/manual entries).
- Send urgent cases back to nurse care when needed.
- Manage tasks, escalations, and discharge summaries.
- Upload clinical documents and review patient history.

Pharmacy

- Review pending prescriptions and dispense medications.
- Track dispensing history and daily stats.
- Handle controlled substances approvals where required.
- Manage inventory, suppliers, purchase orders, and stock adjustments.
- Process refill requests and insurance claims.

Realtime Updates

The realtime server broadcasts events (e.g., new admissions, completed consultations, pharmacy updates) to connected clients. This keeps dashboards in sync without manual refresh.

Support and Troubleshooting

- If login fails, verify the username and password, and ensure the account is active.
- If actions fail, check the API response for error messages and verify database connectivity.
- Admins can review activity logs to trace user actions and errors.